

Affiliated to Pokhara University

POKHARA ENGINEERING COLLEGE

Phirke, Pokhara-8



A

Project Proposal on E-SPEAK

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Submitted to :

Research Management Cell

Pokhara Engineering College

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Chapter 1 Introduction

1.1 Background

In Nepal, civic complaints (like damaged roads, poor sanitation, or broken streetlights) are often reported through informal channels like social media or verbal complaints, leading to inefficiencies in how local governments respond. Our system solves this by offering a smart, location-based platform where citizens can report problems, nearby users can support them, and authorities are alerted when enough people engage making local governance faster, transparent, and more community driven.

1.2 Statement of Problem

These problem statements clearly outline the key issues in the system:

1. Citizens have no structured way to report and track local civic issues.
2. Authorities receive delayed, duplicate, or low-priority complaints.
3. There's no smart system to escalate issues based on public urgency or engagement.

1.3 Objectives

Our project aims to achieve key objectives that include:

1. Develop a Progressive Web Application (PWA) for structured, location-based civic issue reporting with AI-powered classification, sentiment analysis, and duplicate detection.
2. Promote active community engagement through upvoting, commenting, polls, and smart notifications to ensure transparency and participation.
3. Enable efficient government response through automated alerts, real-time dashboards, and data-driven insights, supporting better resource allocation and inclusive governance across urban and rural areas.

1.4 Scope of project

This project will develop PWA for structured, location-based civic issue reporting with AI-powered classification, sentiment analysis, and duplicate detection. It will include community engagement features such as voting and smart notifications to enhance citizen participation. The system will enable automated alerts to authorities based on priority and public feedback.

1.5 Limitations

The project does come with some limitations.

1. The PWA may have limited functionality in offline or low-bandwidth areas.
2. AI may misclassify issues with unclear or incomplete data, especially for local languages.
3. It depends on active participation from both users and government officials.

Chapter 2 Literature Review

AI-Driven Civic Issue Reporting and Engagement Platforms

The rise of Civic Technology (Civic Tech) has transformed citizen-government interactions. Platforms now allow users to report issues (e.g., broken infrastructure, sanitation) and receive updates from authorities. Recently, the integration of Artificial Intelligence (AI) and Natural Language Processing (NLP) has enabled automated classification, urgency detection, and feedback loops (Linders, 2012).

2.1. Early Civic Tech Platforms

Early platforms like **FixMyStreet** and **SeeClickFix** offered basic issue submission and location tagging but required manual sorting of complaints, slowing response times (King & Brown, 2007). Crowd-sourced tools like **CitySourced** and **Ushahidi** showcased the potential of civic participation but lacked automated issue handling (Meier, 2015).

2.2. The Rise of AI and NLP in Civic Applications

AI and NLP now allow platforms to automate key processes: classifying issues, detecting urgency through **sentiment analysis**, and flagging priority cases (Shin & Hwang, 2020; Dey et al., 2021). However, AI models can be biased or inaccurate if not properly trained (Liu et al., 2019), requiring ongoing updates and validation.

2.3. Location-Based and Context-Aware Civic Reporting

Modern systems integrate **GPS and GIS**, generating heatmaps of reported issues and enabling location-based alerts (Batty et al., 2012). However, **privacy** must be protected when using location data (Perng & Kitchin, 2018).

2.4. Feedback Mechanisms and Transparency

Effective platforms provide **status updates** to users, boosting trust and engagement (Nam & Pardo, 2011). Features like **user ratings** and the ability to **reopen issues** help ensure accountability (Linders & Wilson, 2014).

2.5. AI Retraining and Continuous Improvement

AI models must be regularly retrained to stay effective as new issues arise, and user behavior evolves (Mehrabi et al., 2021). Admin tools for retraining classifiers help maintain system relevance.

2.6. Challenges and Considerations

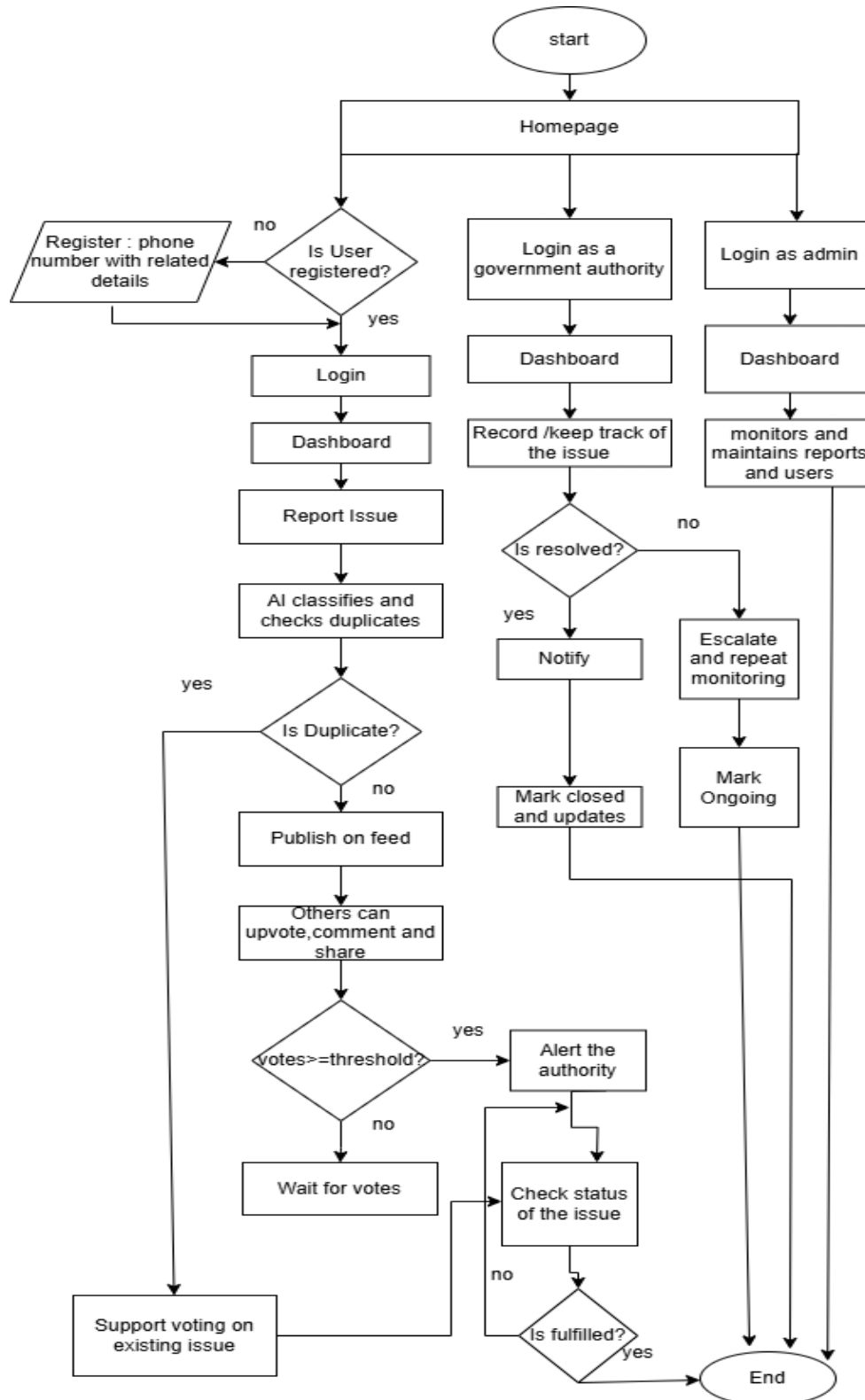
- **Misclassification** can reduce user trust.
- **Bias and fairness** must be monitored (Mehrabi et al., 2021).
- **Privacy** concerns arise from handling location and sentiment data (Zhou et al., 2020).

2.7. Summary

AI and NLP can significantly improve civic issue platforms by automating classification, prioritization, and feedback. For **E-Speak**, combining modular AI engines, transparent communication, and location-aware engagement can enhance responsiveness and citizen participation while challenges of fairness, privacy, and AI accuracy must be carefully addressed.

Chapter 3 Methodology

Flowchart:



Tools to be used

For the dynamic front-end, strong back-end, flexible data handling, AI and NLP we will use the following tools to develop our system and the tools are given below:

1. For frontend:

- a. HTML
- b. CSS
- c. React.JS

2. For backend:

- a. Python Django framework

3. Database

- a. Mysql and PostgreSQL+ Post GIS

4. AI and NLP

TYPE	Tools/Libraries
Natural Language Processing (Text processing and Similarity)	SpaCY, NLTK, Text Blub, IndicNLP(For Nepali)
Classification (AI Models)	scikit-learn
Translation	googletrans
Location Matching	Geopy, Post GIS

Chapter 4 Project Schedule

Milestones	Jun 2025	July 2025	Aug 2025	Sept 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026
	4 th week	1 st - 4 th week	1 st - 4 th week	1 st - 4 th week	1 st - 4 th week	1 st - 4 th week	1 st - 4 th week	1 st - 4 th week
Project Planning								
Literature Survey								
System Designing and modeling								
Front End Implementation								
Back End Implementation								
Documentation								

Fig: Project Schedule

Chapter 5 Expected Outcomes

Our project expected outcomes can be highlighted as below: -

1. Efficient Issue Detection and Reporting:

AI-powered classification (e.g., road damage, garbage, sewage) with NLP support for Nepali and informal text enables accurate and quick reporting.

2. Increased Civic Participation and Awareness:

The platform encourages citizens, especially youth, to actively engage in local governance and increases awareness of civic rights and responsibilities.

3. Strengthened Community Collaboration:

Features like upvoting, commenting, and information sharing foster community-driven problem-solving and collective action for social good.

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To,

Date: 2025-06-23

The coordinator
Research Management Cell
Pokhara Engineering College
Phirke, Pokhara-8

Subject: Request for proposal presentation of final project.

Dear sir,

It is pleasure to inform you that, we Ashbin Rajbomsi(21075356), Dhiren Sen Thakuri(21075361), Santosh Baniya(21075380) Bachelor of Computer Engineering students of this college, are conducting a project entitled “ E-Speak”. In this regard, we would like to request you give permission for proposal presentation of our final project which is under the planned presentation schedule of the college.

Looking forward to your kind cooperation.

Thanking You!!

Name of students: -

Ashbin Rajbomsi, Dhiren Sen Thakuri, Santosh Baniya
Bachelor of Computer Engineering, 7th SEM
Pokhara Engineering College (PEC), Pokhara-8, Phirke

Approval Letter

As per student's application, on behalf of supervisor, The Project group fulfilled all the requirements of college for project work and granted him/her for proposal presentation of final project as mentioned in college schedules.

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Er. Krishna Ghimire
Coordinator, Research Management Cell.